Lesson 8.4 — Sample Statistics

Big idea: A sample statistic (like the sample mean) is our best estimate of a population parameter. The sampling distribution describes how that estimate varies from sample to sample.

Quick review

Sample mean (x■): sum of values divided by sample size n. Estimates μ.
Sample proportion (p■): count of successes divided by n. Estimates p.
Sampling distribution: the distribution of x■ (or p■) over all possible samples of size n.
Standard error of mean: SE(x■) = σ/√n. Standard error of proportion: SE(p■) = √(p(1 – p)/n).
Central Limit Theorem: for large n, the sampling distribution of x■ is approximately normal regardless of the population shape.

Worked example

Worked example. A sample of 100 high schoolers has x■ = 7.2 hours sleep, σ (population SD) ≈ 1.2 hours. Find the standard error of the sample mean.

$SE = \sigma/\sqrt{n} = 1.2/\sqrt{100} = 1.2/10 = 0.12$ hours.

Warm-ups

1. Compute the mean of 4, 7, 10, 11.
2. If 30 of 50 sampled students walk to school, what is the sample proportion?
3. If $\sigma = 6$ and $n = 36$, find the standard error of the sample mean.

Practice

1. A poll of 400 voters finds 240 support a candidate. Find p■ and SE(p■).

2. Sample of $n = 64$ with population $\sigma = 8$. Find SE(x■) and explain how SE changes if n quadruples to 256.

3. Sample heights (in inches): 64, 66, 68, 70, 72. Find the sample mean and (using sample SD ≈ 3.16) estimate SE if you imagine taking similar samples of size 5.

Challenge

1. A bag of marbles is 30% red. You sample 100 marbles. (a) Mean and SE of p■. (b) Approximate the probability that p■ is at most 0.25.

2. If a sample mean has SE = 0.5, how large must n be to halve the SE to 0.25, assuming σ stays the same?

Answer key — Lesson 8.4

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $(4 + 7 + 10 + 11)/4 = 8$.

2. p■ = $30/50 = 0.6$.

3. SE = $6/6 = 1$.

Practice

1. p■ = 0.6. SE = $\sqrt{(0.6 \cdot 0.4 / 400)} = \sqrt{(0.0006)} \approx 0.0245$.

2. SE = $8/8 = 1$. With $n = 256$: SE = $8/16 = 0.5$. Quadrupling n halves the SE (general rule: SE scales with $1/\sqrt{n}$).

3. Mean = 68. Using sample SD as an estimate: SE $\approx 3.16/\sqrt{5} \approx 1.41$ inches.

Challenge

1. (a) Mean of p■ is 0.30. SE = $\sqrt{(0.3 \cdot 0.7 / 100)} = \sqrt{0.0021} \approx 0.0458$. (b) $z = (0.25 - 0.30)/0.0458 \approx -1.09$. P(p■ $\le 0.25) \approx P(Z \le -1.09) \approx 0.138$ (about 13.8%).

2. SE = $\sigma/\sqrt{n}$. To halve SE you need $\sqrt{n}$ to double, so n must quadruple.